



Efficient glycolysis of waste polyethylene terephthalate textiles over Zn-MCM-41 catalysts

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ABSTRACT

Polyethylene terephthalate (PET), also known as polyester, is extensively utilized in the textile industry. The most commonly used chemical degradation method of PET is ethylene glycol (EG) glycolysis, which enables a closed-loop recovery process suitable for large-scale production. In this study, a new type of heterogeneous catalyst (Zn-MCM-41) was proposed for glycolysis of waste PET textiles, and their catalytic performance is much higher than other mesoporous zeolites. Under optimal reaction conditions, over Zn-MCM-41-25 (Si/Al molar ratio of 25) catalyst PET achieved a conversion ratio of 100 %, with bis-hydroxyethyl terephthalate (BHET) yield reaching 81.4 %. XPS results show that zinc is mainly presented as the state of Zn^{2+} , resulting in excellent stability of the catalyst. HPLC and LCMS analyses were conducted on the products obtained from PET glycolysis, revealing that only a small amount of oligomers were generated during the process. These oligomers primarily consisted of dimer $[\text{BHET}]_2$ and trimer $[\text{BHET}]_3$. The Zn-MCM-41-25 catalyst also demonstrates exceptional performance in colored waste PET textiles glycolysis. Furthermore, we investigated the kinetics of PET glycolysis and determined it to follow first-order kinetics with an activation energy value of $210.03 \text{ kJ mol}^{-1}$. Additionally, HPLC methods was established for analyzing glycolysis products and recovering catalysts using a simple filtration technique. The results demonstrate that even after five cycles, the catalyst maintains its high catalytic activity.

1. Introduction

Polyethylene terephthalate (PET), also known as polyester, is extensively utilized in the textile industry due to its favorable physical and chemical properties. Nearly 70 million tons of PET are produced worldwide each year [1], yet the recycling ratio remains significantly low (approximately 10 %) [2]. The accumulation of waste PET in the environment poses a severe impact on both the environment and human health, primarily because of PET's resistance to degradation and inadequate recycling rates. The exploration and establishment of post-consumer recycling initiatives for PET textiles are imperative to achieve closed-loop polyester utilization.

Currently, PET recycling technology can be categorized into three methods: mechanical [3,4], physical [5] and chemical methods [5–8]. While mechanical methods are commonly employed for non-polluting waste PET field recycling, they necessitate advanced sorting technologies resulting in high labor costs and low efficiency. Physical methods are widely used for polyester waste recycling; however, with each subsequent cycle of recycling, the processing performance, stability, and

coloring ability of recycled products tend to deteriorate leading to a decline in quality [6]. Evidently, neither of these two methods alone can achieve the goal of a circular economy for PET. In contrast to mechanical or physical techniques mentioned above, the chemical method involves degrading PET through breaking internal ester bonds within polymer molecules using various reaction media and catalysts. This depolymerization reaction results in corresponding oligomers or monomers [9–11], which can then be repolymerized under specific reaction conditions to produce polyester fibers or other high value-added products, such as dyes, softeners, dye retardant, dye fixing agents and unsaturated polyester resins. [7,12,13]. The recycling of bis-hydroxyethyl terephthalate (BHET) is conducive to improving the limited amounts of resources available on the planet. The chemical method offers greenness, economic viability, and sustainability characteristics for PET degradation [10], making it an essential approach towards addressing environmental issues associated with waste PET. The commonly employed chemical methods for PET degradation include hydrolysis [14–17], alcoholysis, glycolysis [18–20], reductive depolymerizations [21], and catalytic pyrolysis [1,22,23] based on different reaction

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media. Among these methods, PET glycolysis is the most widely studied, due to its mild reaction conditions and high safety. It primarily employs ethylene glycol (EG) as the reaction medium to obtain the monomer product BHET under specific reaction conditions [24–27]. Subsequently, BHET is repolymerized for the production of new PET products, enabling a closed-loop recovery process suitable for large-scale production.

Numerous catalysts have been reported for PET glycolysis, categorized as either homogeneous or heterogeneous based on the phase states of the catalysts and reactants during the reaction process. The main homogeneous catalysts include metal salts [10,28,29], ionic liquids [9,30,31], deep eutectic solvents [20,27,32], and so on. Zinc acetate [26,28] serves as a representative traditional homogeneous catalyst with significant importance in catalyzing PET glycolysis. In 1987, Vaidya and Nadkarni [33] successfully produced polyester polyols from waste PET using zinc acetate as a catalyst. Subsequently, Ghaemy et al. [29] and Cao et al. [10] investigated the catalytic effects of various metal cation acetates on PET glycolysis and concluded that zinc acetate exhibited superior catalytic performance. Homogeneous catalysts offer larger contact areas between the catalyst and reactant resulting in higher activity; however industrial application poses challenges such as separation difficulties, which can affect subsequent recrystallization product yield and purity. The development of an economically viable homogeneous catalyst that enables efficient separation and recyclability poses a formidable challenge, yet it also presents a unique opportunity for innovation within the research community focused on heterogeneous catalysts.

Heterogeneous catalysts mainly include metals, metal oxides [24,34], and composite catalysts. The composite catalysts prepared with zeolite [35], MOFs [36], or silica [37,38] as the carrier supported by metals, metal oxides, ionic liquids, eutectic solvent, etc. Suo et al. [36] discovered for the first time that MOFs have excellent catalytic activity during PET glycolysis. Then, Wang et al. [39] loaded eutectic solvent into ZIF-8 as a composite catalyst for PET glycolysis. After 25 min, complete degradation of PET was achieved with a catalyst loading of 0.4 wt% at 195 °C, and the catalyst could be recycled after 6 cycles with a PET degradation ratio of 100 % and BHET yield close to 80 %. Besides, the heterogeneous catalyst can also be recovered by magnetic methods for magnetic supports. Cano et al. [37] found that paramagnetic ionic liquid-coated $\text{Fe}_3\text{O}_4@(\text{SiO}_2@(\text{mim}))[\text{FeCl}_4]$ catalyst can be used at 180 °C for 24 h after 12 consecutive reaction times without significant loss of catalytic activity (BHET yield 84 %); while additional efforts are needed to lower the reaction time to improve the energy-efficiency. Despite its wide application in chemical industries and its importance in the earth abundance heterogeneous catalysts, zeolites have been less extensively studied for PET glycolysis. In 2008 Shukla et al. [40] proposed to use beta and Y zeolites as catalysts under mild conditions, and the highest BHET yield of 65% was obtained over Y zeolite in 8 h at 196 °C. Recently, Yao et al. [35] compared the glycolysis efficiency of SBA-15 zeolite catalysts supported by different metal oxides and found that the efficiency is in the order of $\text{ZnO} > \text{MgO} > \text{CdO} > \text{CoO} > \text{NiO} > \text{CuO}$, which once again proved the superior catalytic activity of zinc species in PET glycolysis. However, heterogeneous catalysts suffer from poor dispersion properties due to internal diffusion limitations [41], resulting in slightly lower catalytic activity and selectivity compared to homogeneous catalysts. Therefore, finding suitable heterogeneous catalysts is crucial for addressing challenges related to difficult recovery and low utilization of catalysts during industrial degradation of waste PET. Moreover, the activity of most reported catalysts were tested for clear and hot washed PET and the effectiveness for PET with contaminants are unknown.

The present study aimed to design a series of heterogeneous and highly efficient catalysts, Zn-MCM-41 with different Zn contents and Si/Al ratios, for the glycolysis of PET using EG. Firstly, a series of mesoporous zeolites were evaluated as potential catalyst materials, and the Zn^{2+} -modified zeolite catalyst was prepared via an ion exchange

method. Subsequently, the optimal catalyst material was selected based on its performance. Secondly, the glycolysis reaction conditions and kinetic behavior of the catalyst were thoroughly investigated. A liquid phase analysis method was developed for analyzing glycolysis products. Furthermore, a reliable catalyst recovery method was established. Simultaneously, the stability, activity, and acidity of the catalyst during recycling were examined to provide experimental evidence for its industrial application in glycolysis.

2. Experimental section

2.1. Materials

The PET fabric samples were provided by Beijing Sinceyou Engineering Technology Co., Ltd., and subsequently cut into rectangular blocks (approximately 20 mm × 15 mm). Various zeolites (MCM-41, Mesoporous Y, and SBA-15) were obtained from Nankai University Catalyst Co., Ltd. EG (AR, ≥99.5 %) was obtained from Shanghai Titan Scientific Co., Ltd. BHET standard (AR, ≥85 %) was obtained from Shanghai Yien Chemical Technology Co., Ltd. Zinc nitrate (AR, ≥99 %) was obtained from Beijing InnoChem Science & Technology Co., Ltd. Zinc acetate (AR, ≥99 %) was obtained from Shanghai Chemical Industry Co., Ltd. Zinc oxide (AR, ≥99.9 %) was obtained from Beijing MREDA Technology Co., Ltd. Dimethyl sulfoxide (DMSO, AR, ≥99 %) was obtained from Tianjin Xiensi Biochemical Technology Co., Ltd. Methanol (HPLC grades) was obtained from MERYER (Shanghai) Chemical Technology Co., Ltd.

2.2. Catalyst preparation

The zeolite catalysts were prepared using the ion exchange method. The corresponding raw zeolite powder was added to an aqueous solution of zinc nitrate, followed by heating at 90 °C in an ad water bath for 4 h. Subsequently, the mixture was dried through rotary steaming and further dried in an oven at 60 °C for 2 h. Finally, calcination at 550 °C for 300–360 min was performed to obtain the modified zeolite catalyst.

2.3. Glycolysis of PET

In a three-neck flask equipped with a stirring rotor (as shown in Fig. S1), PET fabric/EG/catalyst with a certain proportion (5.0 g/30.0 g/0.25 g) were added. The flask was sealed on the left and right sides with tin foil and then heated in an oil bath. After heating for 30–120 min, hot filtration was performed with 200 mL of deionized water at 90 °C to ensure complete removal of the catalyst and residual reactants from the flask. The resulting filter cake was dried in the oven at 60 °C, and the filtrate was subjected to further treatment by heating it in a water bath at 90 °C for 30 min before rapid filtration. As above-mentioned the filter cake (the oligomers) was dried in a 60 °C oven, while the filtrate was refrigerated at −4 °C for a duration of about 12 h. Subsequently, the refrigerated filtrate underwent filtration to obtain white needle-like crystals BHET, as depicted in Fig. S2. In the glycolysis performance of PET in this study (Section 3.1), the conversion ratio of PET, the yield of BHET, and the selectivity of BHET were calculated based on weight method by the following formula:

$$\text{Conversion ratio of PET} = \frac{m_{\text{PETO}} - m_{\text{PET}}}{m_{\text{PETO}}} \times 100\% \quad (1)$$

$$\text{Yield of BHET} = \frac{m_{\text{BHET}}/M_{\text{BHET}}}{m_{\text{PETO}}/M_{\text{PET}}} \times 100\% \quad (2)$$

$$\text{Selectivity of BHET} = \frac{m_{\text{BHET}}/M_{\text{BHET}}}{(m_{\text{PETO}} - m_{\text{PET}})/M_{\text{PET}}} \times 100\% \quad (3)$$

where m_{PETO} and m_{PET} are the initial and after reaction weight of PET, respectively; m_{BHET} is the weight of the BHET obtained; M_{BHET} (254 g/

mol) and M_{PET} (192 g/mol) are the molar mass of BHET and PET per repeating unit, respectively.

2.4. Catalyst characterization

The X-ray diffraction (XRD) patterns of the powder materials were recorded using a Rigaku-2038 X-ray diffractometer equipped with Cu K α radiation ($\lambda = 1.54056 \text{ \AA}$) and scanning at a rate of $10^\circ/\text{min}$, with an operating voltage of 40 kV and current of 35 mA in the range of 2θ from 5 to 90° . The zeolite catalyst was analyzed by differential scanning calorimetry (DSC) and thermogravimetric analysis (TGA) using an STA 449 F5 instrument. The chemical state of zinc species on the catalyst surface was investigated through X-ray Photoelectron Spectroscopy (XPS), employing Thermo Kalpha X-ray photoelectron spectroscopy. The metal content of the catalyst before and after the reaction was quantified using Agilent ICP-OES 5800, which utilizes inductively coupled plasma chromatography.

2.5. Analysis of reaction products

The obtained BHET crystals and oligomers were analyzed by differential scanning calorimetry (DSC) and thermogravimetry (TGA) on a STA 449 F5 integrated thermal analysis instrument. The heating rate was $10^\circ\text{C}/\text{min}$ within the temperature range of $35 \sim 800^\circ\text{C}$ under nitrogen atmosphere. The BHET products were analyzed by ^1H and ^{13}C NMR spectroscopy on a BRUKER AVANCE 400 nuclear magnetic resonance spectrometer. For the solution containing BHET products, high-performance liquid chromatography (HPLC) analysis was performed using Shimadzu LC-2030 Plus equipped with a UV detector and C18 column. The column temperature was maintained at 35°C employing methanol: water (70 v: 30 v) as the mobile phase with the flow rate of 0.5 mL/min. Each time 10 μL sample was injected, and the total running time was 20 min. Liquid chromatography-mass spectrometry (LCMS) analysis of glycolysis products was performed by Vanquish Flex UHPLC Orbitrap Exploris 240 MS LMS.

2.6. Catalyst recycling

The catalyst recycling experiment (Section 3.3.) used HPLC analysis method to detect the yield of BHET. The method is as follows: after glycolysis, the mixture was cooled to room temperature and subsequently treated with 150 mL of DMSO over a period of 6 h. Then, the supernatant was separated through centrifugation, and the resulting solution samples were analyzed by HPLC analysis. The obtained solid

after centrifugation was then calcined in Muffle furnace at 550°C for 5 h. The regenerated catalyst was reused for subsequent glycolysis reactions, while several fresh catalyst was added to maintain consistent amount of catalyst throughout each glycolysis process.

3. Results and discussion

3.1. Glycolysis performance of PET

Firstly, the glycolysis performance of the synthesized catalysts using zinc nitrate as zinc source loading on different types of zeolites (i.e., SBA-15, MCM-41, Mesoporous Y, and MCM-41-25 with a Si/Al molar ratio of 25), as well as the metal oxide (ZnO) and the homogeneous catalyst $\text{Zn}(\text{OAc})_2$ was measured, and the results are illustrated in Fig. 1a. Furthermore, a comparison with the reported results of traditional homogeneous and heterogeneous catalysts is listed in Table S1. It can be observed that both ZnO-MCM-41 and Zn-MCM-41-25 exhibit significantly higher catalytic efficiency compared with SBA-15 and Mesoporous Y, which is the same as that of the surface area data of the zeolite catalysts, as shown in Fig. 1b. Additionally, it can be seen that the catalytic efficiency of the 5 %ZnO-MCM-41 catalyst is slightly lower than that of $\text{Zn}(\text{OAc})_2$ but higher than that of ZnO alone. The proposed ZnO-MCM-41 and Zn-MCM-41-25 catalysts demonstrate much higher BHET yield compared with that of the traditional heterogeneous catalysts.

The effect of zinc content in ZnO-MCM-41 and Zn-MCM-41-25 catalyst on the glycolysis of PET investigated. When using ZnO-MCM-41 zeolite as the catalyst (Fig. 2a), the PET conversion ratio, selectivity and yield of BHET increased rapidly with increasing zinc loading from 5 % to 13 %. As the Zn content continued to increase from 13 % to 15 %, no further obvious improvement on the catalytic performance can be observed. Similarly, when utilizing Zn-MCM-41-25 zeolite as the catalyst (Fig. 2b), the PET conversion ratio, selectivity and yield of BHET also increased with increasing zinc loading from 1 % to 8 %. However, further increases in zinc loading resulted in a decrease in the conversion ratio, selectivity and yield of BHET due to excessive calcination at high temperatures leading to the formation of ZnO that blocked the pores within the zeolite structure and consequently reduced its specific surface area. The highest catalytic activity was achieved at zinc loading of 8 %.

The metal content in the catalyst before and after the reaction was determined using ICP analysis. The results revealed that the actual zinc loading of 8 % Zn-MCM-41-25 catalyst was found to be 6.53 %. After glycolysis, the content of zinc decreased slight to 6.48 %, indicating a minimal amount of metal shedding during the reaction (0.05 %). On the

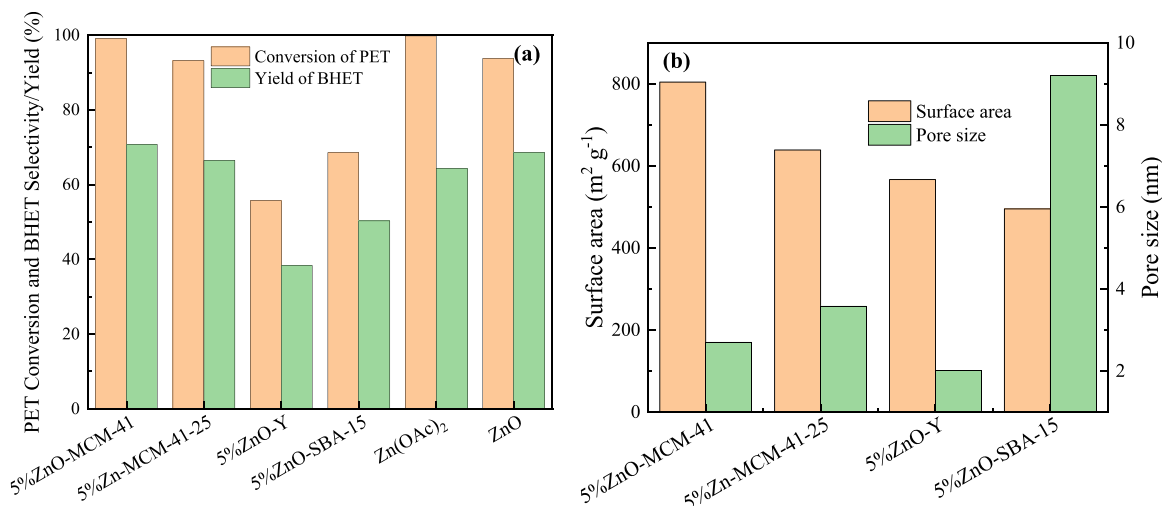


Fig. 1. (a) PET glycolysis over different Zn-based catalysts under the conditions of 5 g PET, 20 g EG, 0.25 g catalyst, and 190°C for 120 min. (b) Pore size and surface area of mesoporous zeolite catalysts.

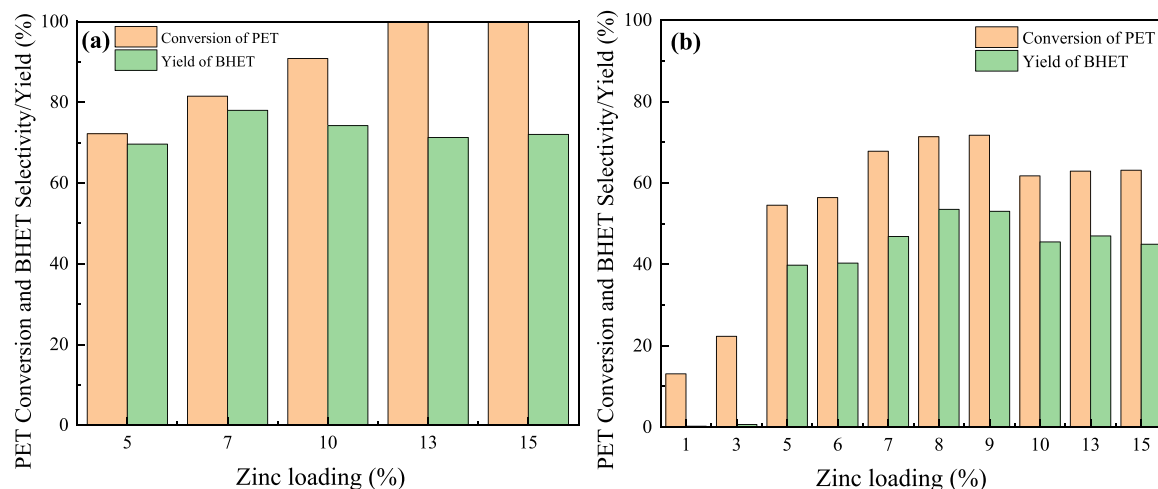


Fig. 2. PET glycolysis with different Zn loading catalysts. a) ZnO-MCM-41 catalyst under the reaction conditions: 5 g PET, 20 g EG, 0.25 g catalyst, 190 °C for 120 min. b) Zn-MCM-41-25 under the reaction conditions: 5 g PET, 20 g EG, 0.05 g catalyst, 190 °C for 120 min.

other hand, for the 15 % ZnO-MCM-41 zeolite catalyst was measured as 13.49 %. However, after glycolysis, there was a significant reduction in zinc content to 12.21 %, suggesting a higher degree of metal shedding during the reaction (1.28 %). This can be attributed to that the all-silicon zeolite MCM-41 has no exchange sites for metal ions, and thus resulting

in weaker attachment of zinc species on its surface that zinc is presence in the form of ZnO. Zn-MCM-41-25 with Si/Al ratio of 25 led to the transform of Zn^{2+} to the skeleton of the structure, behaving as active sites by ion exchange method, and exhibits higher stability in the reaction process. Consequently, Zn-MCM-41-25 zeolite catalyst with zinc

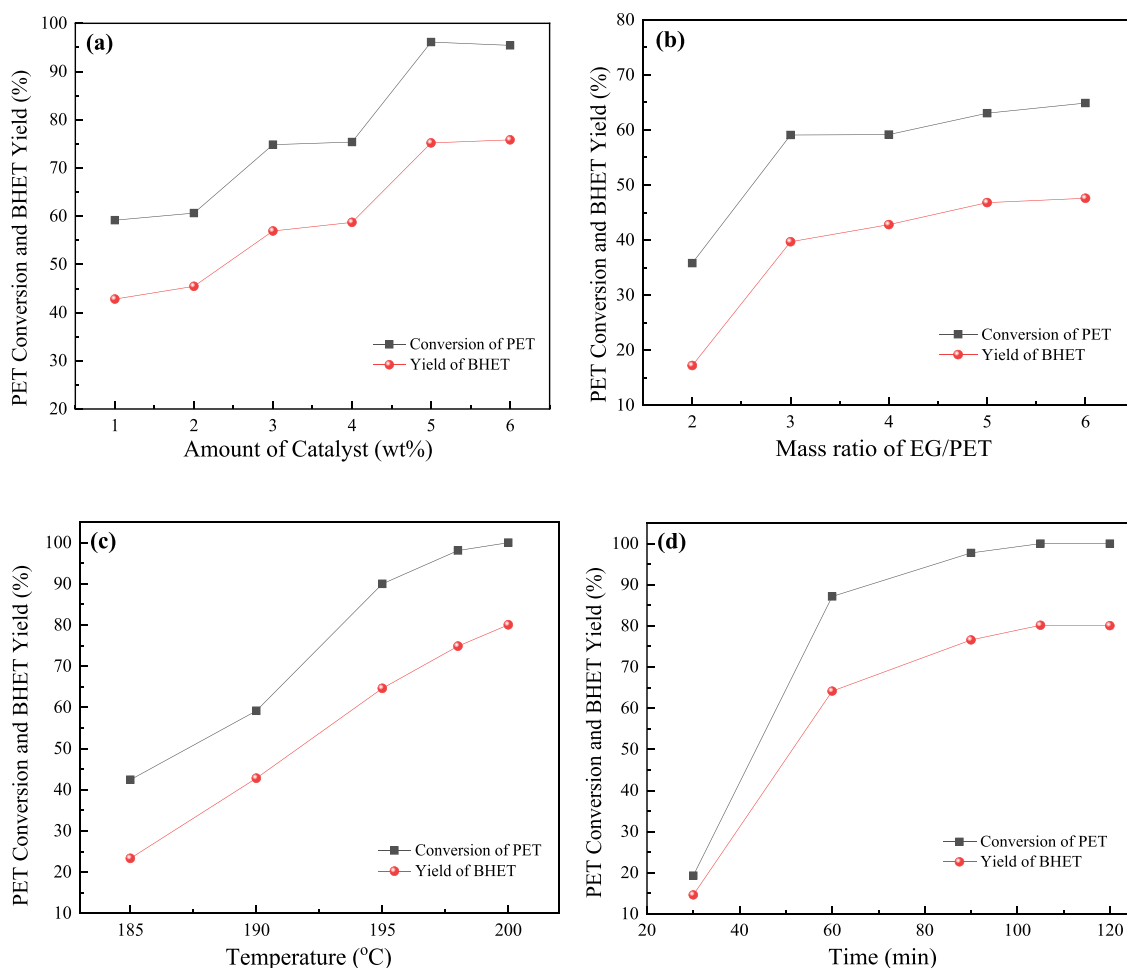


Fig. 3. Effect reaction conditions on PET glycolysis. a) Amount of catalyst effect under the reaction conditions: EG/PET mass ratio of 4 at 190 °C for 120 min. b) Mass ratio of EG/PET effect with catalyst amount of 1 wt% at 190 °C for 120 min. c) Reaction temperature effect with catalyst amount of 1 wt%, EG/PET mass ratio of 4 and reaction time of 120 min. d) Reaction time effect with catalyst amount of 1 wt%, EG/PET mass ratio of 4 at 200 °C.

loading of 8 % was selected for the experiment of optimizing reaction conditions for PET glycolysis.

3.2. Optimization of reaction conditions

To determine the optimal reaction conditions for PET glycolysis over 8 % Zn-MCM-41-25 catalyst, the effects of the amount of catalyst (mass ratio to PET), the mass ratio of EG/PET, reaction temperature and reaction time were investigated, and the results are illustrated in Fig. 3. Under the reaction conditions: reaction time of 120 min, temperature of 190 °C, EG/PET mass ratio of 4, the influence of catalyst amount (ranging from 1 wt% to 6 wt%) on PET glycolysis was investigated (Fig. 3a). The results revealed that when the catalyst amount increased from 1 wt% to 5 wt%, there was a significant increase in PET conversion ratio and BHET selectivity, due to the increase of the catalyst amount providing more active sites for interaction with PET. Further increasing in catalyst amount resulted in no significant change in both PET conversion and BHET yield. Therefore, the optimal catalyst amount was chose as 5 wt%. The EG/PET mass ratio effect on PET glycolysis results shown in Fig. 3b demonstrated a positive correlation between the amount of EG and the PET conversion and BHET yield. This can be attributed to the increased contact area between PET and the reaction medium resulting from an elevated quantity of EG. Consequently, the ester bond in PET macromolecules becomes more susceptible to break, enhancing the PET degradation efficiency. Notably, when the mass ratio of EG/PET is 6, the degradation reaction of PET reaches a state of saturation.

As shown in Fig. 3c, the PET conversion and BHET yield increased significantly with the increase of reaction temperature due to the endothermic nature of depolymerization. At high temperature, the swelling of PET would benefit the chain PET dissolving in EG, and then rapidly diffusing to the surface of the catalyst and interacting with the active center. When the reaction temperature reached 200 °C, PET was basically completely degraded, and the BHET yield reached more than 80 %. Then, the reaction temperature was chosen as 200 °C to further investigate the reaction time effect. The results (Fig. 3d) demonstrated that when the reaction time ranged from 30 to 90 min, the concentration of oligomer and BHET in the solvent was minimal, facilitating a rapid progression of the glycolysis reaction into its positive phase. As the reaction time increased, there was enhanced contact between the catalyst and reactant, leading to a significant improvement in glycolysis efficiency. After 105 min, PET degradation reached near completion with maximum conversion ratio and BHET selectivity. Subsequently, further prolonging the reaction time had negligible impact on PET conversion. In summary, with 8 % Zn-MCM-41-25 as the catalyst for PET glycolysis the optimal reaction conditions are: catalyst amount of 5 % (mass ratio to PET), reaction temperature of 200 °C, EG/PET mass ratio of 6, and reaction time of 105 min, and the PET conversion and BHET yield can achieve 100 % and 81.4 % (results obtained by weight method). Moreover, the operating conditions optimization results for Zn-MCM-41 catalyst are illustrated in Fig. S3, and under the optimal conditions the PET conversion and BHET yield can achieve 96 % and 72 %.

3.3. Kinetic studies on the PET glycolysis

According to previous reports, the kinetic models of PET glycolysis include first-order kinetics [35,36], shrinkage nucleus model [20,42], and other mathematical models [43,44]. It is assumed that the glycolysis reaction follows first-order kinetics, and the kinetic equation is:

$$Kt = \ln \frac{1}{1-x} \quad (4)$$

where K is the reaction rate constant; t is the reaction time; x is the conversion ratio of PET.

To investigate the effect of temperature on PET glycolysis, the

$\ln\left(\frac{1}{1-x}\right)$ as a function of reaction time was plotted in Fig. 4a. The results demonstrate a strong linear relationship with a correlation coefficient greater than 0.9. The activation energy E_a can be calculated by applying the Arrhenius formula:

$$\ln K = -\frac{E_a}{RT} + \ln A \quad (5)$$

where K is the reaction rate constant; E_a represents activation energy of the reaction; R is the ideal gas constant; T is the Kelvin temperature; $\ln A$ represents pre-exponential factor. Using $\ln K$ as Y-axis and reciprocal temperature ($1/T$) as X-axis the activation energy of 210.03 kJ mol⁻¹ was obtained, which is slightly smaller than that of PET glycolysis over ZnO/SBA-15 catalyst (215.91 kJ mol⁻¹) as reported [35]. It demonstrates that the catalyst used in this paper has a good catalytic effect on the glycolysis of PET.

3.4. Catalyst characterization

The MCM-41-25 zeolite was loaded with zinc ions by ion exchange method to obtain the modified catalyst. TGA analysis (Fig. 5a) revealed a weight loss of approximately 2 wt% till 200 °C, primarily attributed to the evaporation of a small amount of crystal water within the catalyst. The weight loss of zinc-modified catalyst is less than that of zeolite raw powder, indicating excellent thermal stability of the synthesized catalyst for PET glycolysis within reaction temperature range of 180 ~ 210 °C.

XRD curves of Zn-MCM-41-25 zeolites with different zinc loadings are shown in Fig. 5b. Amorphous peaks attributed to MCM-41 were observed around 23°. When the loading of Zn is less than 10 %, there is no obvious diffraction peak of zinc metal or metal oxide can be observed in the XRD pattern, indicating uniform distribution of zinc on the zeolite surface and formation of small particles due to the high specific surface area of MCM-41 and low zinc loading.

The chemical state of zinc species on the surface of Zn-MCM-41-25 zeolite catalyst was analyzed by XPS (Fig. 5c). According to literature [45–47], the binding energies for Zn 2p_{3/2} and Zn 2p_{1/2} peaks are reported as 1022 eV and 1044 eV, respectively. The Zn 2p_{3/2} curve can be divided into two components at 1022.1 eV and 1023.3 eV, corresponding to ZnO and Zn²⁺ species, respectively. Based on peak area calculation, it is determined that the content of Zn²⁺ is approximately 59 %. The results indicate that Zn species in the modified MCM-41 zeolite predominantly exist in the form of Zn²⁺, which confirmed that zinc was successfully converted into the structural skeleton via ion exchange method and acted as the active site. Consequently, during catalytic reactions, these zinc species exhibit enhanced stability due to their strong attachment to the catalyst surface.

3.5. Product analysis

In order to determine the chemical structure of the main product, the obtained glycolytic products white needle-like crystals were characterized by ¹H NMR and ¹³C NMR spectroscopy. ¹H NMR characterization (Fig. 6a) revealed that peak No. 4 at the chemical shift of 8.10 ppm corresponds to the four H atoms on the benzene ring. Peaks No. 2 and 1 at the chemical shift of 4.32 ppm and 3.73 ppm represents the four 4 H atoms in methylene -COOCH₂- and methylene of -CH₂-OH, respectively. Additionally, Peak No. 3 at the chemical shift of 4.96 ppm corresponds to two H atoms of hydroxyl-OH. The observed chemical shifts for these aforementioned hydrogen atoms align perfectly with those reported for BHET in literature [10,37,48]. It is worth noting that in an ideal scenario within ¹H NMR spectrum, the ratio between respective areas should correspond directly to the number ratio between corresponding types or positions within their parent substances. In Fig. 6a, it was observed that an area ratio among these four types of H atoms as being approximately equal to: H(benzene):H(methylene-COOCH₂-):H(methylene-CH₂-OH):H

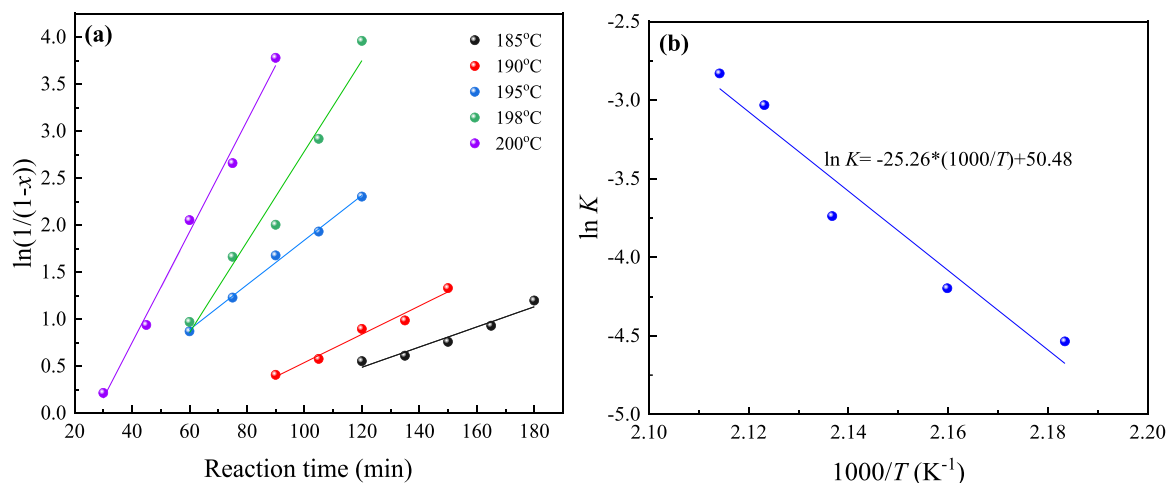


Fig. 4. Kinetics of PET glycolysis. a) Effect of temperature. b) Arrhenius plot.

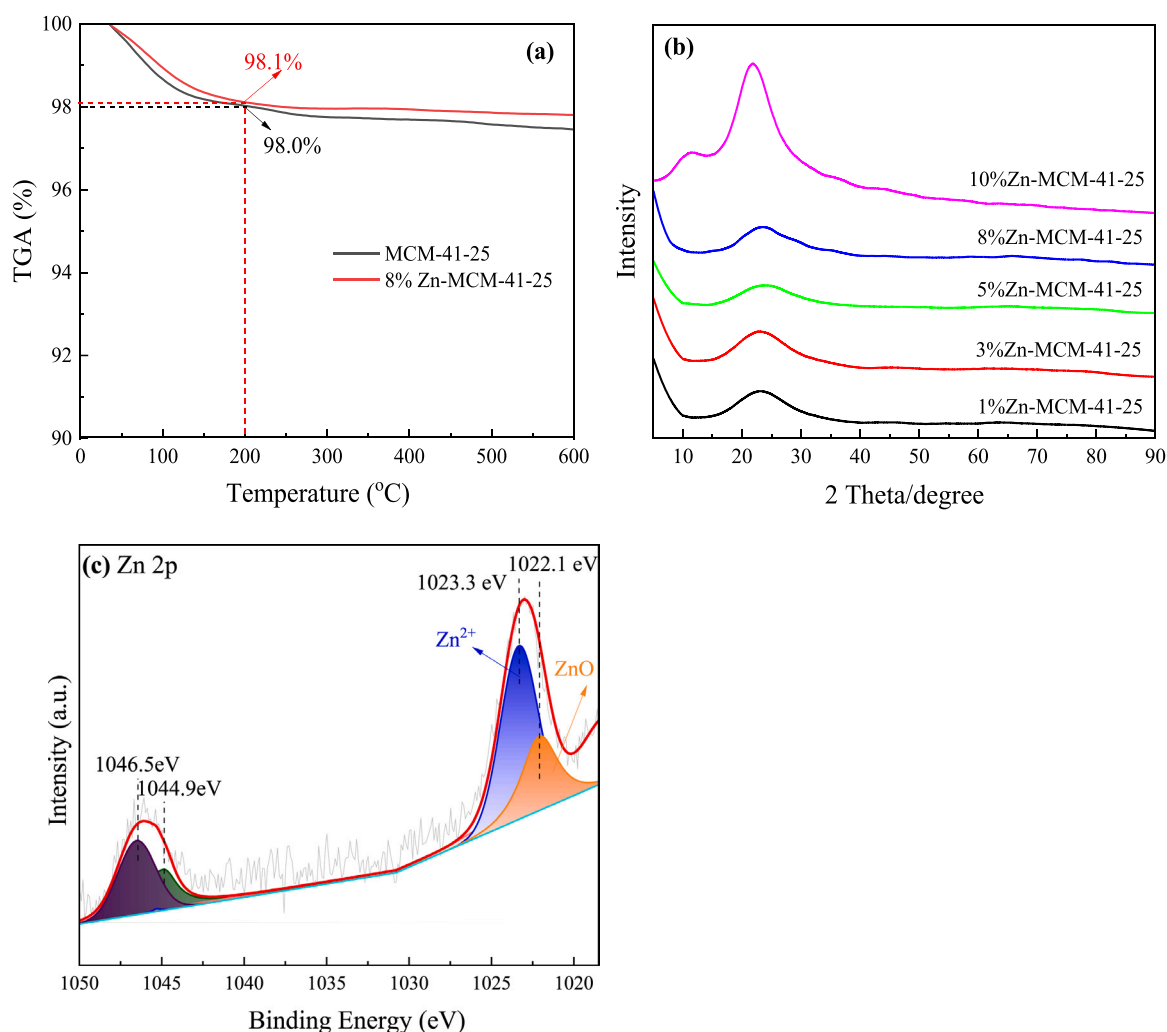


Fig. 5. a) TGA curves of Zn-MCM-41-25 catalyst. b) XRD curves of MCM-41 with different zinc loadings. c) XPS characterization of 8 % Zn-MCM-41-25 catalyst.

(hydroxyl-OH) $\approx$ (2:2:1:2). The corresponding ¹³C NMR spectroscopy result is illustrated in Fig. 6b. The peaks at the chemical shift of 129.48, 133.76, 165.17, 67.02 and 59.05 ppm represent the four C atoms on the benzene ring (Ph), two C atoms on benzene ring of Ph-CO-, two C atoms on -COO-, two C atoms at methylene of -COOCH₂-, and two C atoms at

methylene of -CH₂-OH, respectively. Notably, these chemical shifts align with those reported in literature for BHET [37,49]. Consequently, based on both ¹H NMR and ¹³C NMR results almost no large impurity peaks except for characteristic ones associated with BHET were observed; it can be concluded that high purity is achieved for BHET as the main

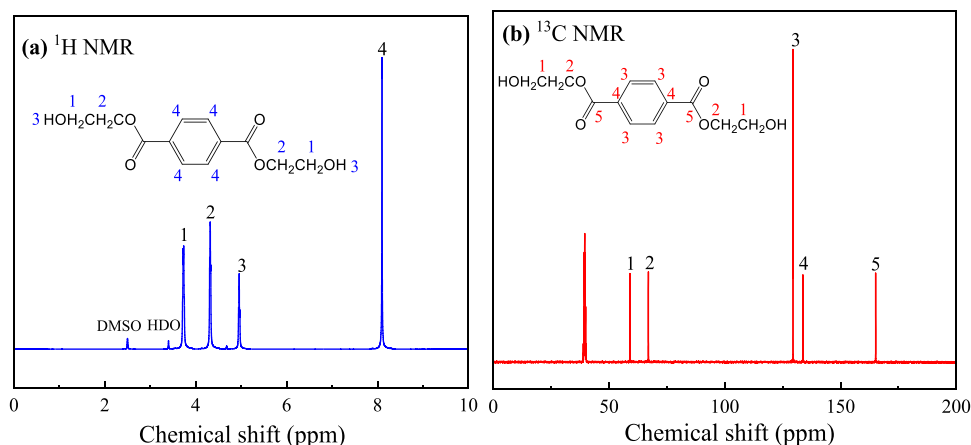


Fig. 6. NMR spectra of BHET product. a) ^1H NMR spectra. b) ^{13}C NMR spectra.

product during PET glycolysis.

The TGA and DTG results of the BHET standard sample (Standard BHET sample), the glycolysis product BHET obtained after filtration (Product sample 1), and the oligomers and BHET mixture obtained without filtration after reaction (Product sample 2), are illustrated in Fig. 7. Consistent TGA and DTG curves among Product sample 1, Product sample 2, and Standard BHET sample indicates minimal presence of incompletely degraded oligomers in glycolysis products and highlight high purity levels achieved for BHET as its main product. The initial weight loss peak occurs at 176.6 °C, accounting for approximately 30.6 % of the total sample weight. This weight loss is attributed to the thermal decomposition of BHET into smaller molecules as temperature increases. Subsequently, a second weight loss begins at 346.5 °C, representing around 55.1 % of the total sample weight. This can be attributed to monomer thermal polymerization leading to oligomer formation at high temperatures, followed by their subsequent decomposition at 346.5 °C [20,50]. When the temperature is higher than 460 °C, no further change in sample mass is observed with increasing temperature. These two weight loss temperatures align well with literature-reported thermal decomposition temperatures for BHET monomer [48,49], confirming that the primary product of this reaction is indeed BHET monomer itself.

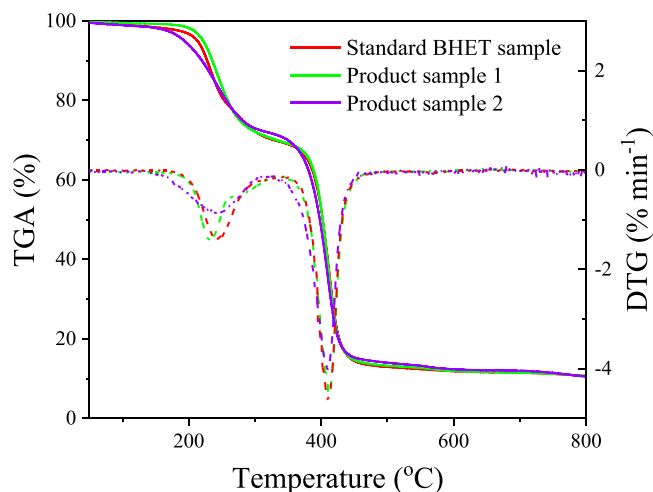


Fig. 7. TGA and DTG curves of Standard BHET sample, Product sample 1 (The glycolytic product BHET was obtained after filtration), and Product sample 2 (The oligomers and BHET mixture obtained without filtration after reaction).

3.6. The glycolysis performance for colored waste PET textile and its reusability

To investigate the glycolysis performance of catalyst for colored waste PET textile and its reusability, the glycolysis products were analyzed by HPLC. As mentioned above the selectivity and yield of BHET for PET glycolysis during the operating conditions optimization studies were determined through the weighting method; however, it should be noted that during the filtration and other processes, there is an inevitable loss of BHET in the product leading to a lower BHET yield. Then, in the following investigations the glycolysis solution after removing the unreacted PET and catalyst was directly refrigerated to obtain BHET and oligomers crystals. The content of BHET in the product was then quantified using HPLC analysis, and the details can be found in Supporting Information. Only monomeric BHET, dimeric $[\text{BHET}]_2$ and trimeric $[\text{BHET}]_3$ species were detected in the HPLC spectra, proving that the only byproducts during PET glycolysis are $[\text{BHET}]_2$ and $[\text{BHET}]_3$.

The reusability of 8 % Zn-MCM-41-25 catalyst for colorless cloth glycolysis was investigated, and the results are illustrated in Fig. 8. It can be observed that the catalyst exhibited excellent recyclability over five cycles, maintaining PET conversion over 95 %, BHET selectivity above 99 %. Subsequently, the proposed catalyst was expanded the application window to waste PET textile with the color army blue glycolysis, as shown in Fig. 8b. It can be seen that the BHET selectivity can achieve a value of 98.7 %, slightly lower than that obtained for colorless cloth glycolysis. The initial PET conversion on the fresh catalyst is comparable to that observed for colorless cloth glycolysis. However, after five cycles the PET conversion exhibit a little decreasing, indicating the exists of macromolecular dyes exhibiting adverse effects on the glycolysis. The ICP analysis was conducted to determine its metal content before and after each cycle. The results revealed that the actual zinc loading in the 8 % Zn-MCM-41-25 zeolite was found to be approximately 6.53 %. After one cycle, the zinc content slightly decreased to 6.48 %, while after five cycles it further reduced to about 4.56 %, resulting in the decrease in PET conversion. The acidity of 8 % Zn-MCM-41-25 zeolite before and after reaction was compared by NH_3 -TPD characterization, and the results are illustrated in Fig. S6. The results proved that the acidity of the catalyst decreased slightly after 5 cycles due to the fall off of ZnO during the reaction, which was consistent with the results of ICP.

3.7. PET glycolysis mechanism

The H^+ on MCM zeolite was replaced by Zn^{2+} through the ion exchange method to form a composite catalyst. Based on the experiment results and literature reports [9–11,20,35], zinc salts typically function as Lewis acid catalysts in the transesterification reaction of PET

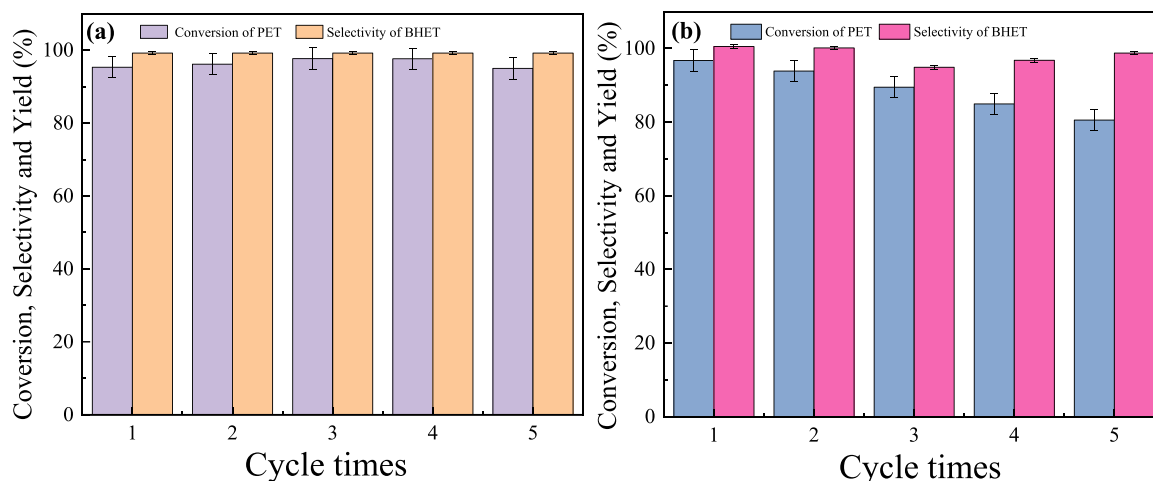


Fig. 8. Recycling of catalysts. a) Colorless cloth. b) Colored cloth.

glycolysis, reacting nucleophilically with the carbonyl oxygen on the PET chain. Zinc ion attacks the carbonyl oxygen in PET, facilitating the removal of lone electron pairs from this oxygen atom and promoting the formation of $^+O=C$ species, as illustrated in Fig. 9. Subsequently, this positive charge is transferred to the carbonyl carbon, resulting in $O-C^+$ formation that facilitates bonding with an O atom from EG. Through C-O bond formation between EG and carbonyl carbon, disruption occurs within the long chain structure of PET while breaking down ester group C-O bonds and converting long-chain PET into short-chain fragments. Additionally, under interaction with Zn^{2+} and carbonyl carbon, O-H bond of EG dissociates into H^+ , which can further attack C=O bond within PET oligomers leading to ester bond cleavage and continued

cracking of these oligomers. Ultimately, polymers are cracked into oligomers that continue undergoing cracking processes until monomer BHET is formed. Finally, the desorption of BHET products facilitates catalyst recycling.

4. Conclusion

The present study focuses on the design and synthesis of a highly efficient heterogeneous catalyst for the conversion of waste colorless and colored PET textiles into BHET. Various zinc-modified mesoporous zeolites (SBA-15, MCM-41-25, MCM-41-Silicon, Mesoporous Y) were prepared via ion exchange method as catalysts for PET glycolysis.

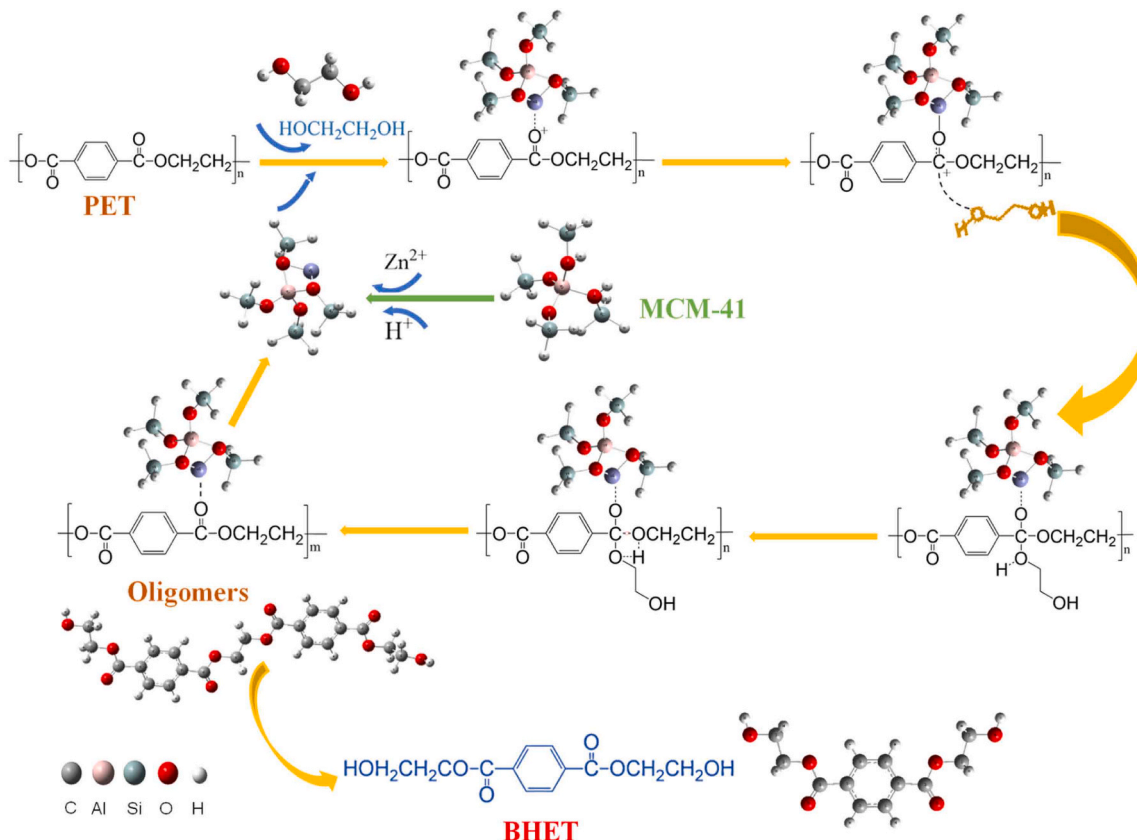


Fig. 9. Proposed PET glycolysis mechanism over Zn-MCM-41-25 catalyst.

Among them, the catalyst based on MCM-41 exhibited superior activity and was selected for further evaluation. The PET glycolysis reaction using Zn-MCM-41-25 catalyst with Si/Al ratio of 25 and 8 % Zn loading capacity was investigated under optimized conditions: catalyst amount of 5 % (mass ratio to PET), reaction temperature of 200 °C, EG/PET mass ratio of 6, and reaction time of 105 min, and the PET conversion and BHET yield can achieve 100 % and 81.4 % (results obtained by weight method). HPLC and LCMS analysis confirmed that only a small amount of oligomers were formed during glycolysis, primarily dimer [BHET]₂ and trimer [BHET]₃. Furthermore, kinetic studies revealed that the glycolysis reaction followed first-order kinetics with an activation energy of 210.03 kJ/mol in the presence of 8 % Zn-MCM-41-25 catalyst. Notably, this catalyst demonstrated excellent stability even after five consecutive uses in PET glycolysis reactions. In conclusion, our findings propose a novel approach to address challenges associated with inefficient recovery and utilization of catalysts in industrial degradation processes involving waste PET materials.

CRedit authorship contribution statement

Zhanpeng Wang: Validation, Data curation. **Yiming Liu:** Investigation, Data curation. **Chengna Dai:** Writing – review & editing, Writing – original draft, Methodology. **Gangqiang Yu:** Writing – review & editing, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

No data was used for the research described in the article.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.cattod.2024.114827](https://doi.org/10.1016/j.cattod.2024.114827).

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